

Figure 5: Qav traffic shaping

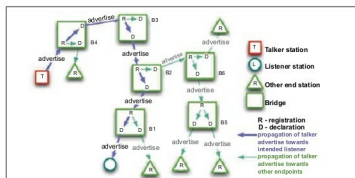


Figure 6: Successful reservation (talker advertise)

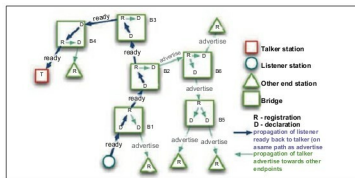


Figure 7: Reservation acknowledged (listener ready)

of the hardware; otherwise (if the value of X is negative), the AVB packet is buffered until the value of X rises again and becomes positive. During this interval, the hardware gets busy dealing with non-AVB traffic, thus handling AVB and non-AVB traffic efficiently.

This is just a short summary of how the hardware reserves bandwidth for AV traffic, to ensure the desired QoS. Refer Figure 5.

IEEE 802.1Qat

The stream reservation protocol ensures the reservation of network resources for specific AV streams traversing bridged networks. It determines the resources required, and exploits a mechanism for dynamic maintenance of those resources.

Before the source starts initiating the AV traffic, it sends the 'talker advertise' message across the network. This message consists of the following: source and destination MAC address, 16-bit unique ID and information about the resource

requirements. The bridge receiving this message sets aside or reserves the requested resources, and propagates the message to the next system. The listener responds with a 'listener ready' message, which is sent back to the talker. If any bridge fails to allocate the requested resources, it sends a 'talker fail' message to the other systems, which results in AV traffic streaming getting aborted. This is how a stream is actually registered with the bridges and the two endpoints – talker and listener – communicate with each other. The resources comprise ensuring port throughput, data rate and registering AVB class. Refer Figure 6 and Figure 7.

IEEE 1722 Layer 2 AVB transport protocol

This protocol basically handles the packetising of the AV data into a standard 1722 Ethernet frame. It also shoulders the responsibility of passing down the 1722 AVB packet to the driver for it to transmit over the Ethernet.

This protocol is implemented as part of the talker application, which does the following:

- Reads raw audio/video data from a file
- Packetises the same data into a standard AVB Ethernet frame
- Delivers the AVB Ethernet frame to the lower layer driver for its transmission

This application has also undergone significant modifications, which involve reading data directly from the AV devices and streaming it to the listener application. The main purpose of this is to extract the raw data and play it in real-time over the loudspeaker/headphones, or when using an ALSA sub-system.

We are currently experimenting with an AVB set-up that involves AVB endpoints (talker/listener) designed and developed by XMOS and an AVB Ethernet switch from DSP4U. Initially, our set-up included only one talker and one listener, with support from the first three AVB protocols, but now the AVB stack has evolved and is mature enough to handle multiple AVB devices in an AVB network. We will discuss the AVDECC, MAAP and IEEE 1722 layer 2 transport protocols in the next article about multiple AVB devices in an AVB network.

But before we do that, here's some food for thought. Can you think of another domain where this technology can penetrate? What will an entire AVB network set up using Ethernet LAN cables look like? Quite clumsy, right? So what's next? An AVB wireless chip? 

References

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- <http://www.ieee802.org/1/pages/avbridges.html>
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Videos

- <https://www.youtube.com/watch?v=R8HjyU4llkE>
- <http://www.xmos.com/news/videos?search=AVB>
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